Pintu Lal **Meena**

Staff Software Engineer | Platform, AI & Distributed Systems

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Summary

Staff Software Engineer and Chief Architect with 8+ years of experience designing distributed platforms, payment systems, and Generative AI architectures. I do not just build isolated solutions; I architect for extensibility and the next 100X scale by identifying patterns and developing optimized, reusable components. At Amazon, this philosophy drove the transformation of fragmented legacy services into a unified, plugin-based platform and the creation of a storage-agnostic caching framework , which collectively reduced new feature launch effort by 50%. Whether automating deployment strategies for peak traffic events or achieving modular firmware reuse across IoT fleets, I prioritize system efficiency, maintainability, and accelerated development cycles. Currently, I apply these principles to build production-grade, multi-modal AI pipelines.

Work Experience

YehiGo Technologies (Proprietary AI Engine) — *India*

Founder & Chief Architect — *2024 - Present*

- **Generative AI Platform Architecture :** Built Kineti, a production-grade Generative AI platform delivering story driven automated text-to-video creation.
- **Autonomous Research-to-Script Intelligence :** Designed an agentic pipeline that performs live internet search for a given topic, synthesizes information from multiple external documents, and generates structured, long-form scripts for downstream video generation.
- **Multimodal Model Orchestration :** Developed proprietary prompt-engineering and orchestration logic integrating Qwen and LLaMA (text LLMs), Stable Diffusion and LoRAs (visual generation), and neural audio engines (Kokoro, Coqui) into a unified multimedia pipeline.
- **Scalable, Secure Cloud Infrastructure :** Designed a cloud-native, queue-driven microservices architecture for compute-intensive workloads and implemented end-to-end encrypted P2P communication using the Signal Protocol to ensure efficiency, privacy, and user trust.

Amazon — *Hyderabad, India*

Software Engineer II (Technical Lead, SDE3 Scope) — *Feb. 2019 - Jul. 2024*

- **Federated Platform Transformation(Org-Wide) :** Spearheaded the evolution of a fragmented service with ambiguous ownership and responsibilities into a unified, multi-tenant, plugin-based platform. Defined clear ownership boundaries and built a **"safety-net"** instrumentation layer providing end-to-end customer journey visibility and high-fidelity BI insights and automatically monitors feature health, ensuring 100% visibility.
- **Technical Leadership & Business Impact :** Led **6** cross-functional teams, mentoring **12+** engineers, eliminating ownership gaps and cross-team dependencies. Reduced new feature launch effort by **50%**, accelerated client onboarding, and improved collaboration for **100+** stakeholders, delivering a **7%** increase in Conversion, **1.8%** reduction in user aborts (Scan-&-Pay), and a **7.28%** increase in ASR (Send Money).
- **End-to-End Performance, Security & Access Control :** Drove deep optimizations across backend, Android, and iOS layers, achieving a **90%** reduction in P90 latency (**5s →540ms** on Android; **85%** improvement on iOS). Streamlined authentication and access-control paths, improving platform security, efficiency, and user experience at scale.
- **Distributed Systems & Scalable Caching :** Executed the migration of high-volume monolithic services to an AWS-hosted microservices architecture , improving P90 latency by **53%**. Designed a robust, storage-agnostic caching framework using the Decorator design pattern, supporting L1/L2 levels and automated pluggable update strategies.
- **Scalability & Deployment Automation :** Architected safe, Full Continuous Deployment (CD) pipelines across distributed services. Templatized a data-driven auto-scaling strategy for high-traffic peak events (IPL, Prime Day, Diwali), eliminating manual dependency calculations, saving days of engineering effort, and educating teams on the new automated workflow.
- **Engineering & Operational Excellence :** Standardized service monitoring metrics and pioneered auto-dashboarding solutions, drastically reducing on-call triage time and enabling rapid identification of impacting components. Revamped infrastructure for independent, parallel testing, removing major developer bottlenecks and manual QA effort.
- **Security, Compliance & Long-Term Strategy :** Implemented high-security storage for iOS devices with advanced data protection. Authored and presented a multi-year technical strategy to org leadership, driving adoption of a new tech stack. Served as POC for regulatory audits and automated stress testing and CI/CD pipelines.

PicoStone — *Mumbai, India*

Software Engineer — *Mar. 2017 - Aug. 2018*

- **Embedded Systems & ML-Driven Device Intelligence :** Designed a semi-supervised learning approach for AC remote training, accelerating model convergence by **62.5%** while reducing required training data by **87.5%**.
- **OTA Firmware & Fleet Reliability :** Architected and deployed a secure Over-The-Air **(OTA)** firmware update system with centralized control, enabling safe rollout to **1,000+** devices with **99.99%** update reliability.
- **Time-Series Data Integrity & Energy Analytics :** Implemented interpolation-based recovery for missing energy telemetry to reflect near-real energy usage, improving reporting accuracy and reliability for customer-facing analytics.
- **Adaptive Data Compression & Query Performance :** Designed a multi-resolution data retention strategy, automatically down-sampling historical energy data (per-second →per-minute/hour) based on dashboard time range. Prevented browser crashes on large queries while preserving analytical fidelity and improving frontend performance.
- **Platform Ownership & Systems Leadership :** Led a **3-person** engineering team to design and ship embedded IoT products (Polar, IRStone) on FreeRTOS, achieving **≈50%** data compression, modular firmware reuse, and consistent **99.99%** device availability across the fleet.
- **End-to-End Delivery (Embedded + Backend) :** Owned the full lifecycle across embedded firmware and backend services, including data pipelines, server-side load distribution, testing, and production rollout.

Education

IIT Bombay (Indian Institute of Technology Bombay) *Mumbai, India*

B.Tech in Computer Science and Engineering *Jul. 2012 - Aug. 2017*

Skills

- **Distributed Systems & Platform Engineering →** System Design, Microservices, Distributed Caching, Observability, Performance Optimization, Authentication & Access Control, API Design, System Architecture
- **AI & Data Systems →** Agentic Workflows, Multi-modal LLM Orchestration (LLaMA, Qwen), RAG, Diffusion Models, Time-Series Data Processing
- **Cloud & Infrastructure →** AWS (EC2, SQS, Lambda, API Gateway), CloudFormation, CI/CD, Monitoring
- **Languages & Foundations →** C++, Java, Python, SQL | Data Structures, Operating Systems, DBMS, OOP

Open Source & Technical Ecosystem

- **Author | Credit Management Library →** Architected a production-ready, database-agnostic AI billing engine featuring a thread-safe credit reservation model (via contextvars) and a dual-write ledger system to prevent LLM overcharging; averages 100 downloads/week
- **Contributor | Eel Framework →** Optimized the core Eel desktop GUI library by implementing native Microsoft Edge App Mode and legacy IE rendering support.
- **Author | YeahDown | Desktop Media Downloader →** Engineered a lightweight, multi-threaded Windows desktop environment utility to download media from web video sources for non-technical users, 80+ monthly downloads on SourceForge.
- **Author | ScriptExtractor | Developer Productivity Tool →** Architected and published a code isolation tool on the VS Code Marketplace that automates the parsing, extraction, and clean separation of tightly coupled inline JavaScript from HTML files, accelerating frontend refactoring workflows.
- **Author | Real-Time Multi-Player Game Engine →** Designed a stateful, fault-tolerant gaming architecture featuring an automated server-side state machine that seamlessly substitutes disconnected players with autonomous robot logic to ensure 100% session continuity.
- **Contributor | playing-cards Ecosystem →** Spearheaded a major repository refactor to decouple core business logic from UI rendering, eliminating a heavy Flutter SDK dependency from backend deployments while maintaining 100% backward compatibility.